

SUREN GOURAPURA

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EDUCATION

Princeton University Ph.D. in Physics, 3.68 GPA

Completed all degree requirements, formally awarded in Sept Aug 2026

Masters in Physics Apr 2021

The Ohio State University Bachelors of Science, 3.94 GPA, with Honors

Majors: Physics, Astronomy, Minor: Mathematics May 2019

AWARDS AND HONORS

- Antarctica Service Medal 2023

RESEARCH EXPERIENCE

(i) SPIDER Experiment: Jul 2019 - Aug 2019

Prof. William Jones, Princeton University:

Pointing Reconstruction for SPIDER-2:

- Improved SPIDER-2 pointing with a time-varying model derived from CMB intensity fluctuations
- Characterization of pointing errors using compact source statistics

SPIDER-2 low-level analysis:

- Development of data quality flags and statistics
- Determination of primary calibration and instrument parameters through CMB cross-correlation

SPIDER-2 integration and testing campaigns:

- Winter 2022-23 deployment campaign of SPIDER-2 in McMurdo, Antarctica
- Summer 2021 integration campaign of SPIDER-2 in Palestine, TX
- Spring 2020 integration and operation of the cryostat in Princeton, NJ

Development of spectral domain component separation pipeline:

- Created unique divisions of the SPIDER sky and published evidence of spatial variation

SPIDER-2 hardware development:

- Mechanical design, testing, and assembly of flight photovoltaic arrays
- Development and implementation of flight data storage system
- Investigation of efficient, cryogenic filters for mitigation of RFI

(ii) Bayesian Cosmological Data Analysis Course: Sept 2024

Cosmoglobe group, University of Oslo: 2 weeks

- Applied SPIDER-1 data at the map level to Commander 3, an end-to-end Bayesian parameter estimator that combines data from multiple experiments

(iii) High Energy Neutrino Astronomy: Jul 2016 - May 2019

Laboratory of Prof. Amy Connolly, The Ohio State University

SURF calibration with genetic algorithms:

- Wrote a genetic algorithm to calibrate SURF data from SLAC

Genetically Evolving NEuTrIno teleScopes (GENETIS) antenna design:

- Evolved wire antennas using gain and neutrino shower simulation software

ARA hardware, firmware, and integration:

- Development of ARA DAQ and Power boxes

ARA antenna modeling summer internship:

- Fitted spherical-harmonic coefficients to generate directional gain patterns
- Built two ARA DAQ stations and refurbished two more, all deployed at the South Pole

(iv) Big Data Analytics Project: May 2018 - Aug 2018

Prof. Richard Hughes and Prof. Brian Winer, The Ohio State University

- Developed a new class, “Big Data” Analytics in Physics 6820, with three other students. Documented here: www.github.com/SummerBigData
- Tackled two Kaggle open problems with machine learning models

PUBLICATIONS

Selected works

- P. A. R. Ade, et al. “Analysis of Polarized Dust Emission Using Data from the First Flight of SPIDER” (2025) ApJ, arXiv 2407.20982
- Julie Rolla, et al. “Evolving Antennas for Ultra-High Energy Neutrino Detection” (2020) PoS, arXiv 2005.07772
- Philippe Voyer, et al. “Multi-sensor fusion for fine-guidance and milliarcsecond-level attitude estimation of balloon-borne telescope” (2026), arXiv 2607.28816
- Simon Tartakovsky, et al. “picoMUX: microcontroller based time-domain multiplexing readout for kilopixel TES arrays” (2026), arXiv 2607.13004
- Jared L. May, et al. “Sub-Kelvin Cryogenics for a Super-Pressure Balloon-Borne CMB Polarimeter: Taurus” (2026), arXiv 2606.29485
- Elle C. Shaw, et al. “In-flight performance of Spider’s 280-GHz receivers” (2024) JATIS, arXiv 2408.10444
- Alexandre E. Adler, et al. “Modeling optical systematics for the Taurus CMB experiment” (2024) JCAP, arXiv 2406.11992
- Jared L. May, et al. “Instrument overview of Taurus: a balloon-borne CMB and dust polarization experiment” (2024) Proc. SPIE 13094, *Ground-based and Airborne Telescopes X*; 1309432, arXiv 2407.01438
- Simon Tartakovsky, et al. “Thermal architecture for a cryogenic super-pressure balloon payload: design and development of the Taurus flight cryostat” (2024) Proc. SPIE 13094, *Ground-based and Airborne Telescopes X*; 130944B, arXiv 2410.18150
- Julie Rolla, et al. “Evolving Antennas for Ultra-High Energy Neutrino Detection” (2021) PoS, arXiv 2112.00197
- Sankar Renu, et al. “Surface engineered polyanhydride-based oral Salmonella subunit nanovaccine for poultry” (2018) Int. J. Nanomed.

TALKS

- **SPIDER & Taurus Mtg:** Talks on Pointing, Mapmaking, Beams, and Pol Convention Jun 2026
- **Pan-Experiment Galactic Science Group:** Polarized Dust Emission with SPIDER Jan 2025
- **SPIDER & Taurus Mtg:** Cross-spectra + Deprojection with Planck May 2024
- **CWRU Cosmo Group:** Foregrounds with SPIDER: Examining Deviations in Dust Modeling Dec 2023
- **The Scientific Committee on Antarctic Research, AAA:** Foregrounds with SPIDER: Examining Deviations in Dust Modeling Sept 2023
- **American Physical Society, Denver CO:** Evolving Antennas for Ultra-High Energy Neutrino Detection (Abstract: R08.00005) Apr 2019

SERVICE AND OUTREACH

- Volunteered on Grad Recruiting for Princeton Phys Dept’s Equity Diversity Initiative Jun 2021 - May 2022
- Developed and hired as an Ambassador in the Princeton Phys Dept’s EDI program Jun 2021 - May 2022
- Volunteered and coordinated high school research opportunities for women Jun 2018 - Jun 2019
- Volunteered for Achieving in Science through Phys. Instr., Research, and Exploration (ASPIRE) Jun 2017 - Jul 2017

ADDITIONAL SKILLS

- Surface-mount soldering, cable making, machining with mill and lathe
- Experience designing with Fusion 360 and Solidworks
- Experience with C++, Python, Bash, Mathematica, and LaTeX
- Woodworking hobby: designed and built tables, stools, shoe racks, benches, a bedframe, and more
- Build my own desktop computers. Knowledgeable about computer parts and assembly